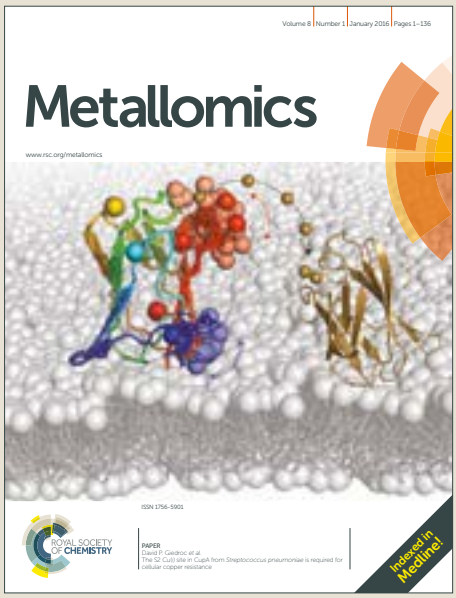
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Cellular Copper Homeostasis: Current Concepts on its interplay with glutathione homeostasis and its implication in Physiology and Human Diseases

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Abstract

Copper is a trace element essential for almost all living organisms. But level of intracellular copper needs to be tightly regulated. Dysregulation of cellular copper homeostasis leading to various diseases demonstrate the importance of this tight regulation. Copper homeostasis is regulated not only within the cell but also within individual intracellular compartments. Inactivation of export machinery results in excess copper being redistributed into various intracellular organelles. Recent evidences suggest involvement of glutathione to play important role in regulating copper entry and intracellular copper homeostasis. Therefore interplay of both homeostasis might play important role within the cell. Similar to copper, glutathione balance is tightly regulated within individual cellular compartments. The review explores the existing literature on the role of glutathione in regulating cellular copper homeostasis. On one hand, interplay of glutathione and copper homeostasis performs important role in normal physiological processes, example neuronal differentiation. On the other hand, perturbation of the interplay might play key role in the pathogenesis of copper homeostasis disorders.

Introduction

Copper is an essential cofactor for important enzymes like Tyrosinase, Dopamine ß hydroxylase, Peptidylglycine ∞ hydroxylating Monooxygenase and several other enzymes, nicely collated in a review (1). The role of copper dependent proteins have been implicated in neurotransmission (2), myelination (3), pigmentation (4) and various physiological and developmental processes (5). Recently it is demonstrated to play an important role in neuronal differentiation (6, 7). Although essential, its level within the cell needs to be tightly regulated. This fact is supported by the occurrence of various diseases caused due to perturbation of copper homeostasis (8, 9). Copper can spontaneously recycle between Cu(I) and Cu(II). Under normal circumstances reducing intracellular environment will result into maintenance of copper mainly in Cu(I) condition (10). But copper accumulation due to inactivation of the export pathway can exceed this cellular buffering capacity, thereby generating oxidative stress (11). Recent

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studies demonstrate that cellular copper and glutathione homeostasis are closely connected.

Cellular organelles have designated chaperones, transporters and proteins for copper delivery, incorporation and maintenance of copper level (12), suggesting that copper homeostasis is tightly regulated in various cellular compartments. Mitochondria and cytosol have exclusive as well as common partners designated to maintain copper level and incorporate copper into copper dependent enzymes. Similarly studies indicate that glutathione homeostasis is also tightly regulated within distinct cellular compartments (13-15).

Existing reviews have nicely collated genetic, biochemical and structural studies to understand various aspects of cellular copper homeostasis and related disorders (8, 16-21). This review intends to explore how copper and glutathione homeostasis is tightly regulated within cellular compartments. It highlights the role of glutathione on copper homeostasis and its implications in normal physiology. It also explores evidences on how changes in cellular copper level affect glutathione homeostasis and how it underlies pathology development in copper homeostasis disorders.

Copper homeostasis in different cellular compartments and related disorders

In mammalian cells copper enters through high affinity copper transporter, Ctr1 (22-24). After entering into the cells, it binds with several copper chaperones, (ATOX1, CCS, COX17), responsible for delivering copper into distinct intracellular compartments (Fig 1). ATOX1 delivers copper to the P(1B)-type ATPases ATP7A and ATP7B (25), present on the membrane of the trans Golgi network (Fig 1). These transporters are responsible for delivering copper into the lumen of the trans Golgi network whereby, copper is incorporated into several cupro proteins (secretory pathway). In response to high intracellular copper, ATP7A and ATP7B localize to endocytic vesicles, towards apical and basolateral membranes respectively (26). The vesicles ultimately fuse with the plasma membrane, thereby exporting excess copper out of the cell (27) (Fig 1). Copper

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chaperone CCS (Copper Chaperone for Superoxide Dismutase) aids in maturation and copper incorporation into Cu-Zn Superoxide Dismuatse1 in the cytosol (28). CCS mediated maturation and copper incorporation into SOD1 also occurs in the mitochondria intermembrane space (IMS) and respective SOD1 are partitioned in the respective compartments (29-31). Therefore CCS has dual role in copper delivery to SOD1 in both cytosol and mitochondria IMS. Another copper chaperone, COX17 participates in copper delivery and maturation of Cytochrome C Oxidase (COX) and plays important role in mitochondrial copper homeostasis (32-34) (Fig 1). Other chaperones for COX maturation (through copper incorporation) include COX11, SCO1 and SCO2 (inner mitochondria membrane) (Fig 2), which will be discussed in detail in the subsequent sections. Chaperones SCO1 and SCO2 are responsible for copper incorporation and maturation of Cu_A site in Cytochrome C Oxidase COX (35) (Fig 2). COX11 is responsible for copper incorporation into the Cu_B sites in (COX) (36, 37). Recently, it has been shown that copper participates not only in maturation, but also biogenesis of COX. Mitochondria resident protein Coa6 is required for biogenesis of the COX2 subunit of COX (38). Exogenous copper supplementation rescuing COX biogenesis defects in coa6Δ *S cerevisiae* links the essential role of mitochondria copper in the process. Again, ATOX1 is responsible for delivering copper to ATP7A and ATP7B, designated to deliver it to the secretory pathway. Thus cellular compartments have distinct molecular partners for maintaining copper homeostasis. This compartmentalization suggests that Copper has distinct essential roles to play in distinct compartments.

Despite the existing compartmentalization, there is efficient communication between organelles to restore homeostasis. Leary and his colleagues nicely demonstrated that SCO1 and SCO2 deficient patient fibroblasts not only have impaired COX activity, but also exhibit copper deprived condition mediated through ATP7A (39). Their study further showed COX19 as a communicating molecule, transmitting this change in mitochondria copper homeostasis to the Golgi based copper transporter ATP7A. This suggests communication of the

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copper homeostases between the mitochondria and the secretory/export pathway. But the exact carrier for exchangeable copper in the cells is still not quite understood. Moreover, while the role of cytosolic and mitochondrial copper can be appreciated, role of copper in the nucleus still remains to be investigated.

Inactivation of cellular copper homeostasis can lead to disruption of vital cellular functions as is evident from the presence of several human diseases caused by disruption of copper metabolism. Inactivation of ATP7A leads to failure of copper being delivered to the blood stream creating systemic copper deficiency called Menkes Disease (MNK) (40, 41). Apart from this, mutations in ATP7A is linked to Occipital Horn Syndrome (OHS) (42). Genetic mutation, inactivating ATP7B, leads to failure of cellular copper excretion resulting into excess copper accumulation in liver, brain and other organs called Wilson Disease (WD) (43-47). WD is characterized by hepatic and neurological symptoms (47). Not only transporters, mutations in the protein partners aiding in the vesicular trafficking of ATP7A and ATP7B have been implicated in copper homeostasis disorders. MEDNIK (Mental retardation, Enteropathy, Deafness, Neuropathy, Ichthyosis, and Keratodermia) is caused by mutations in AP1S1, a clathrin adaptor protein involved in vesicular trafficking of ATP7A and ATP7B (48). Hypocupremia, hypoceruloplasminemia and liver copper accumulation, along with intrahepatic cholestasis are symptoms of this disease (48). Microvillous Inclusion Disease (MVID), an early onset disease affecting at birth to two months age (49), is caused due to mutations in MyosinVb (MyoVb) (50-53). Recent studies have discovered that myosinVb regulates copper mediated movement of ATP7B vesicles from apical compartment to the plasma membrane in polarized hepatocytes (54). This suggests that mutations in myosinVb will affect ATP7B mediated apical copper export. Consequently copper accumulation might be one of the symptoms in MVID patients. Whether patients with MVID exhibit changes in cellular copper homeostasis remains to be investigated. However, deficiency of copper and zinc have been reported in MVID patients (55).

Defects in mitochondrial copper homeostasis contribute to a wide range of disorders. *SCO1* mutation leading to COX deficiency results into neonatal-onset

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hepatic failure and encephalopathy (56). *SCO2* mutations have been identified in patients with infantile cardioencephalopathy (57) and hypertrophic cardiomyopathy (58). Newly described *Coa6* mutation has been identified in mitochondrial cardiomyopathy (59). This underlines the importance of copper not only in the cytosol, but also within the mitochondria.

WD, MEDNIK and possibly MVID are disorders of copper accumulation. Pathologies related to these diseases are due to excess copper affecting important cellular functions. But copper level is regulated not only within the cell, but also within individual cellular compartments. It is therefore important to know the dynamics of copper accumulation in different organelles as a result of disruption of copper homeostasis at different steps. Moreover how does such accumulation affect the compartments, what is the most affected compartment and whether or not there is any inter compartment communication are some of the important questions, which need to be addressed. This is because global copper chelation might alleviate symptoms temporarily but cannot be the permanent strategy for disease management (60). Moreover strategies like gene therapy are still at experimental level, with major challenges like delivery, sustenance and affordability. Liver transplantation in case of WD is performed at the terminal stages and has its own challenges. Understanding the immediate effects of copper accumulation, dissecting the most affected cellular compartments might provide targeted therapy with fewer side effects. For example ATP7B$^{-/-}$ mice have dysregulated Liver X Receptor/Retinoid X Receptor (LXR/RXR) (nuclear receptor) heterodimer (61) and decreased FXR (Farnesoid X Receptor) binding to its target DNA (62). Treatment of ATP7B$^{-/-}$ mice with LXR agonist ameliorates disease progression even under conditions of high copper (61).

Dynamics of copper accumulation due to disrupted copper homeostasis in mouse and cell based model

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Animal models of copper homeostasis disorders have provided important means of studying mechanisms of pathology development. Understanding the dynamics of copper accumulation due to disrupted export machinery and its immediate effects underlies disease mechanism.

Toxic milk mouse (*tx*) (63-65), Jackson tx mouse (tx^{j}) (66), Long-Evans Cinnamon rat (LEC rat) (67, 68) and $ATP7B^{-/-}$ mouse (69) are the recognized model systems for investigating mode and effects of copper accumulation in different cell and tissue types. Tx mouse has Met1356Val substitution in the proposed Transmembrane segment (TM8) of *atp7b,* the murine homologue (63). LEC rat has deletion in the 3' region of *atp7b* (68). Both the models accumulate high copper in the liver. But the exact dynamics of copper accumulation in different intracellular compartments have not been investigated. $ATP7B^{-/-}$ mouse has been generated by inserting an early termination codon in exon 2 of the gene. $ATP7B^{-/-}$ mice (69) accumulate hepatic copper as early as 5-6 weeks. Copper not only accumulates in the cytosol, but also in the nucleus (70, 71) accompanied by marked morphological and molecular changes. High nuclear copper in these mice, suggests that, inactivation of the export machinery results in excess intracellular copper being redistributed to other membrane-encapsulated organelles. It would be interesting to investigate level of copper in those organelles specially mitochondria in these mice. Copper accumulation is associated with molecular changes, for example, expression of Ki-67, a marker for cell proliferation (70). Transcripts related to cell cycle machinery and lipid metabolism are mostly affected in these mice (72). $ATP7B^{-/-}$ mice exhibit dysregulation of nuclear receptors (61, 62). Nuclear receptor agonists are potential targets for therapy, as already described. Copper accumulation in these mice is paralleled with change in the abundance of proteins involve in RNA processing (73) in the nucleus. Increased expression of glutathione peroxidase like Selenoprotein H (73) in the nucleus, suggests potential effects of copper accumulation on redox homeostasis. The evidences suggest nucleus as the affected organelle as well as a target for therapy in $ATP\&B^{-/-}$ mice. Similarly

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accumulation of copper in the nucleus, in addition to cytosol, has been observed in the ATOX$^{-/-}$ Mouse Embryonic Fibroblasts (MEF) cells (74).

A recent study has demonstrated that inactivation of export pathway (inactivation of ATP7A) leads to copper accumulation in nucleus and mitochondria in Menkes Disease Fibroblast cells (11).

The evidences suggest that inactivation of copper export pathway leads to changes in level of copper in all major membrane bound organelles supporting the presence of inter-organelle communication. How is excess copper redistributed from one compartment to another needs to be investigated. The exact form of intracellular copper is still unknown. But urinary copper has been found in association with an unidentified Small Copper Carrier molecule (SCC, 2 KDa) in ATP7B$^{-/-}$ mice (75). SCC is also suggested to be the serum copper carrier. Copper bound to SCC can compete with free copper in uptake by CTR1 (75) suggesting it as a source of exchangeable copper. This form of copper, in excess, might be responsible for the downstream effects on cell proliferation (72), RNA processing machinery (73) in events of disrupted copper export pathway. But the exact form of SCC is still not understood. Does excess of exchangeable intracellular copper affect glutathione balance is still a question. As already mentioned, studies in mice suggest that copper might affect glutathione homeostasis (73). This is a plausible hypothesis as glutathione is the first acceptor of Copper after it enters the cell before it binds to the chaperones (76).

Cellular glutathione homeostasis and its compartmentalization

Glutathione couple (GSH/GSSG) is the major cellular antioxidant defense system. It (GSH) is a tripeptide, L-glutamyl-L-cysteinyl-glycine (77, 78). Synthesis of GSH is an ATP dependant two-step process of which, the first one is rate limiting. At first, Glutamate and cysteine is converted to γ-glutamylcysteine by γ-glutamylcysteinesynthetase. Further γ-glutamylcysteine and glycine is converted to glutathione by GSH synthase (79). Among the total cellular pool, majority of

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glutathione resides in the cytosol and mitochondria (80) suggesting maintenance of its homeostasis specifically important in these two compartments. It participates in various cellular redox reactions, (i) detoxification of electrophiles (ii) Maintenance of thiol status of proteins (iii) efficient running of vital cellular functions (iv) maturation of iron – sulphur clusters (81) (v) as a co factor in conjugation reactions catalyzed by Glutathione transferases in xenobiotic detoxification. It also has role in nitric oxide homeostasis (79) and post translational modification of proteins (82). It can either directly participate in redox reactions or act as a co substrate for enzymes such as glutathione peroxidase (Fig 3) (cytosol) and catalase (peroxisomes) (83). During these processes, GSH is reduced to GSSG (oxidized glutathione). GSSG is converted back to GSH by NADPH catalyzed by Glutathione Reductase (GR) (Fig 3). Maintenance of glutathione ratio involves proper rate of its production, conversion to GSSG and GR catalyzed reduction of GSSG to two molar equivalents of GSH. Maintenance of proper glutathione ratio (GSH/GSSG) is a pre requisite for all the essential reactions involving glutathione.

GSH is synthesized in the cytosol and transported to the nucleus, mitochondria and ER where it has distinct role to perform (83).

Glutathione homeostasis in different cellular organelles

As already mentioned that mitochondria have the highest concentration of glutathione after cytosol. Mn-Superoxide dismutase converts $O_2^{\cdot-}$ to hydrogen peroxide. Removal of hydrogen peroxide relies on the availability of GSH by glutathione peroxidases (GPXs) (Fig 3). Among them GPX1, the major isoform, is found in the mitochondria matrix (14) and cytosol. Membrane associated GPX4, in the mitochondria intermembrane space, protects membrane lipids from oxidative damage (14). In addition to GPX, peroxiredoxin (PRX) also reduces H_2O_2 and in turn is regulated by Thioredoxin (Fig 3). It is probable that

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mitochondria have a higher GSH requirement, to remove excess H_2O_2 generated from $O2^{-\cdot}$ by Superoxide dismutase. $O2^{-}$ generated from Oxidative phosphorylation also adds to the oxidative burden in the mitochondria. Glutaredoxin isoform 2 (GRX2) localizes in the mitochondria matrix and nucleus and isoform 1 (GRX1) in the cytosol. These are GSH-dependant thioldisulphideoxidoreductase, responsible for regulating oxidation of accessible thiols in proteins (84, 85) (Fig 3). Glutathione, synthesized in the cytosol, passes through mitochondria outer membrane through porin channels (86). It cannot traverse the inner membrane because of negative membrane potential due to oxidative phosphorylation. It involves carrier-mediated transport located on the inner membrane of the mitochondria (86). Although transported from the cytosol, oxidized glutathione needs to be reduced within mitochondria for maintaining optimum ratio within the organelle. Compartmentalization of GSH is further evidenced by the fact that depletion of only cytosolic GSH is not enough to cause death of isolated rat hepatocytes. Hepatocyte damage and lipid peroxidation is observed only after depletion of mitochondria GSH (87).

Similar to mitochondria, GSH is transported to the Endoplasmic Reticulum (ER) whereas GSSG cannot be transported back to the cytosol (88). Using ER targeted roGFP sensors, E_{GSH} in ER has been demonstrated to be more oxidizing than E_{GSH} in the cytosol (89). Di sulphide bond isomersation catalyzed by Protein DisulphideIsomerase in the ER, an important step in protein folding, requires glutathione (13).

Although it is assumed that like other cellular compartments, nucleus has its own GSH pool, the precise role of nuclear glutathione remains to be investigated. But keeping in mind its important role in Telomerase activity (90), reduction of cysteines in Zn finger binding proteins (15), DNA synthesis (15) and cell proliferation and differentiation (91, 92), nuclear GSH might be of immense importance. High concentration of GSH in the nucleus during early growth phases of cultured cells, sequestration of GSH in the nucleus during G1 and S-

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phases of cell cycle, support importance and compartmentalization of GSH in the organelle (92, 93).

Cellular glutathione and Copper homeostasis

The mechanism, by which, excess intracellular copper affects cellular processes leading to WD has been an important question. It is speculated that excess copper can inflict oxidative damage (94). But the precise mechanism by which excess copper can create oxidative damage is not known. Whether oxidative damage involves perturbation of the cellular redox couple like glutathione and/or thioredoxin needs to be addressed. Moreover the major carrier of exchangeable copper in the cell is to be identified. Although there is no direct answer to these questions, evidences suggest that GSH can bind intracellular copper. Cellular GSH plays role not only in copper entry but also in cellular efflux.

In Hepatoma cell lines, greater that 60% of cytosolic copper is bound to GSH (95). Cu^{67} uptake experiments, in these cells, suggest copper binding to glutathione before being sequestered by Methallothionein (MT). Copper binding to GSH, after its entry is demonstrated also in HEK293T cells (76). Cellular GSH plays important role in copper uptake by Ctr1 in HEK293T cells. Depletion of cytosolic copper chaperones ATOX1 and CCS has no detectable effect on copper entry. But cellular glutathione depletion decreases cellular copper uptake (76). These results suggest that GSH is the first acceptor of copper after its entry into the cell, following which it is either delivered to the chaperones or sequestered by metallothionein. Therefore it is probable that excess intracellular copper can perturb glutathione balance even before being sequestered.

Not only copper entry, glutathione has important role in regulating redox condition of the cysteines in the N-terminus of copper transporters ATP7A and ATP7B. Copper binding cysteines at the N-terminus of ATP7B remain glutathionylated in cells (96). GRX1 interacts with ATP7A and ATP7B to de glutathionylate the N-terminal cysteines, a pre requisite for copper binding to the ATPases. This demonstrates interaction of GSH with the Golgi based copper transporter ATP7A (Fig 4).

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Depletion of cellular glutathione, by BSO (L-Buthionine-Sulphoximine) treatment, affects vesicular trafficking of ATP7A and leads to intracellular copper accumulation (96, 97). Thus glutathione level not only regulates copper entry but also copper efflux. This makes glutathione a key regulator of cellular copper homeostasis. Not only mammalian cells, such evidence also exists in bacteria. Bacteria lacking copper ATPase CopA, are sensitive to glutathione lowering (98) demonstrating important role of GSH in survival under conditions of elevated intracellular copper. Similar results have been observed in mammalian cells. Mouse fibroblasts lacking copper chaperone ATOX1 ($ATOX1^{-/-}$) are also sensitive to glutathione depletion caused by BSO treatment (99). Glutathione depletion causes copper accumulation but the reverse might not be true, under physiological circumstances. This is because reducing intracellular environment might be able to counteract excess copper (10) created by inactivated copper export pathway. However this might hold true for the cytosolic compartment. But organelles like mitochondria might be more vulnerable (11).

Not only the transporter, cellular glutathione has also been implicated to regulate copper binding by cytosolic copper chaperone ATOX1. Cellular GSH/GSSG ratio regulates the redox status of the copper binding CXXC motif of ATOX1 (99). ATOX1 has three cysteines, two of which reside in the copper binding CXXC motif. Cellular glutathione (GSH/GSSG pair) is responsible for maintaining the cysteine residues in reduced state thereby aiding copper binding (Fig 4). However this process presumably requires GRX1 under low glutathione condition (99). This study did not identify glutathionylated ATOX1 probably due to its transient nature (99). Interestingly interaction of glutathione with the N-terminus of ATP7A is potentially stable and detectable (96). Conformation of the N-terminus (100, 101) along with its interaction with the cytosolic domains (102) might dictate the stability of its interaction with GSH. Copper binding CXXC motif resides in an exposed loop of ATOX1. Hence interaction of ATOX1 with GSH might be transient because of its exposure to the cytosolic milieu. However the hypothesis needs more experimental validation.

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Glutathione is important for mitochondria copper homeostasis. Copper incorporation and maturation of Cu_A and Cu_B sites in COX is an important step in mitochondria respiratory chain. Both processes are redox regulated. Copper incorporation into Cu_A site relies on the redox status of the cysteine thiols of SCO1 and SCO2. However there are 2 hypotheses in the field, both demonstrating the importance of redox regulation in the process (34, 35, 103). According to Leary and his colleagues, SCO2-COX-SCO1 forms a ternary complex. Copper is then sequentially transferred to COX from SCO2 followed by SCO1 (35). Following copper transfer, SCO2 acts as an oxidoreductase regulating the redox status of the cysteines in SCO1 (oxidized) post copper transfer. This aids copper incorporation into SCO1 by COX17 (35) (Fig 2B) for the next cycle. This mechanism proposes that SCO1 can accept copper from COX17 only when its cysteines are oxidized. But Banci and her colleagues proposed that COX17 can donate copper to both oxidized and reduced SCO1 but only to reduced SCO2 (34). Copper delivery, by COX17, to oxidized SCO1 is coupled with transfer of two electrons. Following copper delivery, COX17 is converted to partially oxidized apo-form by mitochondrial GSH (34) (Fig 2A). They proposed that copper delivery to the Cu_A site is performed only by SCO1. SCO2 acts as an oxidoreductase maintaining the cysteines of Cu_A in reduced condition thereby aiding copper delivery by SCO1 (103).

Copper delivery to the CuB site of COX by COX11 is also indirectly redox regulated. COX19 protects Cys^{111} of COX11 from oxidation, a pre requisite for copper delivery by COX11 (104) (Fig 2A). Therefore COX maturation is dependent on redox status of Cysteines in both COX as well as the proteins aiding in its maturation..

Cellular GSH plays important role in copper entry, binding and cellular efflux. It has role in maturation of enzymes participating in the mitochondria respiratory chain. Glutathione and glutaredoxin plays role in copper binding to cytosolic chaperone ATOX1 and copper delivery to the N-terminus of ATP7A/7B. Therefore it can be concluded that proper utilization of copper into several cellular organelles is aided by glutathione and glutathione regulated proteins

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(GRX). Hence proper interplay of copper and glutathione homeostasis might be important for the normal physiology of the individual organelles. This in turn will dictate proper functioning of the cell, tissue and organ systems.

Changes in the interplay of copper and redox homeostasis in normal Physiology

Redox homeostasis has been implicated in various physiological processes like differentiation, cell cycle progression (92) etc. But the role of the interplay of redox and metal homeostasis in physiological processes has been speculated, but not clearly demonstrated. How such interplay is co ordinated within cellular compartments is still unexplored. Perturbations in such interplay can be the major contributory factor towards copper homeostasis disorders (94).

Copper is essential for neuronal differentiation is demonstrated by the occurrence of Menkes Disease characterized by copper deficiency and primarily neurological symptoms. ATP7A and copper dependant Lysyl Oxidase is essential for Zebrafish notochord development (5, 105). ATP7A expression in axons during synaptogenesis supports role of copper homeostasis in neuronal development (106, 107). PC12 cells show increase in copper level and metallothionein (MT-3) upon differentiation (6). Increase in intracellular copper level, with differentiation, is hypothesized to meet the increasing demand of copper into the secretory pathway (7). Copper needs to be incorporated into key enzymes involved in neuronal differentiation. One of them is dopamine-ß-hydroxylase, participating in the biosynthesis of norepinephrine. Peptidyl-glycine-∞-monooxygenase (PAM), required for maturation of neuropeptides, is a copper dependant enzyme. Apart from these two examples, the precise role of copper homeostasis is neuronal development is still not quite understood. What are the essential pathways triggered by copper and how do they vary in different regions of the brain might provide insights to the role of copper homeostasis in the neurons. Further it will aid in understanding mechanism of neuronal pathology associated with not only copper metabolism disorders (Menkes and Wilson Disease) but also Alzheimers Disease (108, 109), Parkinsonism (110) and prion disease (111, 112). But it is

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unlikely that copper homeostasis alone can play the key role in neuronal differentiation. It might be a player in the entire team of players driving neuronal differentiation. Alternatively it might be a part of yet unidentified pathway having key role in neuronal differentiation.

In addition to copper, glutathione homeostasis changes have been a key factor towards neuronal differentiation. Glutathione depletion enhances Nerve Growth factor mediated differentiation of PC12 cells in culture (113). Such changes in cellular glutathione can trigger pathways essential for differentiation.

It is probable that changes in intracellular glutathione can trigger changes in cellular copper at the onset of the differentiation process. Recently it has been demonstrated in chick embryo and neuroblastoma (SH-SY5Y) cells, that glutathione ratio (GSH/GSSG) increase as motor neurons/ neuronal cells differentiate (7). Change in redox environment aid copper incorporation into the secretory pathway, which is required for maturation/activation of proteins/pathways involved in differentiation. This study provides the basic evidence of interplay of the two homeostases in neuronal differentiation. Exact molecular mechanisms triggered by the interplay needs to be investigated further.

Preliminary studies on PC12 cells demonstrate that copper induced oxidation of serotonin can be toxic to the cells through generation of reactive oxygen species (114). The study might not provide any direct information on the effects of the changes in the interplay of both pathways on differentiation. But it highlights the effect of excess copper towards redox stress in the neuronal precursor cells.

Inactivation of Copper homeostasis affecting cellular glutathione homeostasis

It has been speculated for sometime that Copper can participate in Fenton reaction for its redox recycling. Whether such redox cycling can occur in the reducing environment of the cytosol, under normal circumstances, is debatable (10). Evidence suggests that intracellular copper binds to GSH, before binding to the chaperones or sequestered by metallothionein. Therefore, excess copper,

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due to inactivation of export machinery, can perturb cellular glutathione balance (11). Since cellular organelles have distinct glutathione ratio, it would be interesting to evaluate how excess copper affects glutathione ratio in individual organelles. But direct evidence of copper accumulation affecting glutathione balance is still lacking in the mouse models of copper metabolism disorders. However, ATP7B$^{-/-}$ mice have increased abundance of Selenoprotein H in the liver nuclei (73). Selenopritein H, with thioredoxin-like CXXU motif, has glutathione peroxidase activity (115). Nuclear abundance of Selenoprotein H in these mice suggests peroxide accumulation. Peroxide accumulation can be the effect of change in glutathione balance as a result of excess copper accumulation. Alternatively excess copper can generate superoxides thereby increasing the load of peroxide. Nevertheless copper accumulation, suggestively in these mice, affects nuclear redox homeostasis. ATP7B$^{-/-}$ mice have upregulated expression of single isoform of hnRNPA2/B1 (73). Changes in the RNA processing machinery, might be the effect of excess copper or downstream oxidative changes. Cells can regulate spliceosome machinery in response to oxidative, genotoxic and other stress factors (116-118).

Copper accumulation due to inactivation of ATP7A in Menkes Disease fibroblast (both immortalized and primary) affects glutathione balance in the mitochondria (Fig 5) (11). It results into accumulation of hydrogen peroxide. Although copper accumulates in all major intracellular compartments, in these cells, effect of excess copper on redox homeostasis is observed specifically in the mitochondria. It is interesting that nuclear glutathione balance remains largely unaffected from the redox perturbation inspite of high copper. This suggests presence of protective mechanisms operative in this compartment to counteract copper induced changes in glutathione balance (11). Menkes Fibroblast cells show downregulation of transcipts involved in apoptosis, that are regulated by mitochondria peroxide level. Changes in gene expression machinery suggest efficient communication between the mitochondria and the nuclear compartment thereby affecting important cellular processes (Fig 5).

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Leary and his colleagues have observed such communication of copper homeostasis between compartments supporting the concept of intercompartment communication as a result of changes in copper level. They observed that SCO1 and SCO2 patient fibroblasts not only exhibit deficiency of COX activity, but also cellular copper deficiency due to ATP7A mediated copper efflux (39). Translocation of COX19 between mitochondria and cytosolic compartment mediates this communication between mitochondria and the Golgi based efflux pathway (39). It would be interesting to investigate the mechanism of such communication.

It is interesting how excess copper, resulting from inactivation of export pathway, affects different compartments in different cell types. While mitochondria are the site of redox changes in Menkes Fibroblast (11), evidences suggest nucleus to be the affected organelle in $ATP7B^{-/-}$ mice (73). Understanding such differential changes might underlie the effects that excess copper mediates in various organs. Liver and brain, the most affected organs in WD, comprise of multiple cell types. It is important to understand how excess copper in WD affects each of these cell types. How are such effects communicated between organelles and between cells in a specific organ. This can contribute, to a large extent, towards pathology development in WD.

Conclusion

The review brings to focus the interplay of cellular copper and redox homeostasis and its role in normal physiology and diseases. Not only within the cell, interplay of both homeostasis is important within specific cellular compartments. Change in one compartment has important implication on the other. For example change in glutathione balance in the cytosol presumably aides in copper delivery to the secretory pathway during neuronal differentiation. Similarly perturbation in one compartment can also inflict change in another compartment leading to pathology. For example, copper accumulation due to inactivation of Golgi based copper transporter, ATP7A, in Menkes Disease Fibroblast, affects redox homeostasis specifically in the mitochondria. Again, redox changes in the

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mitochondria are transmitted to the nucleus resulting into changes in the gene expression machinery (Fig 5). Hence it is necessary to evaluate the contribution of the redox perturbation as a result of changes in copper level towards pathology development in copper homeostasis disorders. Moreover such perturbations need to be evaluated in specific cellular compartments. Not only that, communication between compartments as a result of perturbation in one, needs to be investigated. Hence to understand the importance of copper homeostasis, one must consider its connection with cellular glutathione homeostasis. Studying such interplay within individual cellular compartments will provide better understanding of the role of copper in various cellular processes and copper homeostasis disorders.

Acknowledgements: This work is supported by Ramanujan Fellowship (SB/S2/RJN-106/2015) and Extramural Research (individual centric) Funding (EMR/2016/003293), Science and Engineering Research Board (SERB), Department of Science and Technology to A. B.

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Legends To Figures

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Fig 1: Cellular copper homeostasis: an overview. After entering the cell through copper transporter CTR1, copper (orange sphere) binds with glutathione. Copper is then transferred to several intracellular chaperones like ATOX1 (purple rectangle), COX17 (green hemishere) and CCS (shaded diamond). CCS delivers copper to SOD1 both in the cytosol and mitochondria intermembrane space (dotted line and arrow). ATP7A/7B (red cylinder), residing on TGN, receive copper from ATOX1. ATP7A/7B traffic to vesicles (spheres containing ATP7A/7B) under high intracellular copper, thereby exporting excess copper out of the cell. Cellular glutathione (GSSG, oxidized –blue ring, GSH, reduced – blue half ring) has been demonstrated.

Fig 2: Mitochondria copper homeostasis highlighting the importance of glutathione in the process. *Panel A*: Cox11, SCO1, SCO2 are involved in metallation of copper centres (Cu_A and Cu_B) in Cytochrome C Oxidase (COX). Partially oxidized Cox17 (green hemisphere), with two S-S bonds and two reduced cysteines transfers copper to Cox11, SCO1, SCO2. Cox17 can transfer copper to both reduced and oxidized SCO1. Copper delivery to oxidized SCO1 is coupled with transfer of electrons resulting into reduction and subsequent metallation. Unlike SCO1, SCO2 can only receive copper from COX17 in reduced form (34). Oxidized COX17 is reduced by GSH. SCO2 is responsible for maintaining the Cysteines in the Cu_A site in reduced form, which facilitates copper binding. SCO1, on the other hand, is responsible for copper inorporation by loop recognition (103). Metallation of CU_B is done by Cox 11. Active COX 11 contains three Cysteines, of which two are oxidized and one is reduced. The third cysteine is protected from oxidation by Cox 19 (104). *Panel B:* Alternatively it has been hypothesized that SCO1, SCO2 and COX form a ternary complex. Copper is sequentially delivered by SCO2 followed by SCO1. Additionally SCO2 acts as an oxidoreductase (dotted arrow) manintaining apo SCO1 in oxidized condition (35) (dotted arrow).

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Fig 3: Cellular redox homeostasis : an overview: Glutathione (GSH) and thioredoxins (Trxs) plays major part in perpetuating intracellular thiol-redox equilibrium. Peroxiredoxins (PRX) and Glutathione peroxidases (GPXs) reduce H_2O_2 to H_2O and they form a transient oxidized Sulfenic acid (Cys-OH) intermediate, which is finally oxidized. Oxidized peroxiredoxin is reduced by thioredoxin (TRX). Oxidized thioredoxin is reduced by thioredoxin reductase (TRXR) using electron donor NADPH. On the other hand, Oxidized GPX is reduced by GSH, which in turn gets oxidized to GSSG. Glutathione reductase (GR) reduces GSSG to GSH utilizing electron donor NADPH. Another major function of GSH in the cell is to reduce glutaredoxin (GRX). GRX is the major redox enzyme responsible for maintaining thiol groups of proteins in reduced form using GSH as a co factor.

Fig 4: Role of glutathione in copper entry and copper binding to ATOX1 and ATP7A. GSH is the first acceptor of intracellular copper after its entry through CTR1 (76). ATOX1 coordinates Cu(I) via a CXXC binding motif . Cellular GSH/GSSG plays major role in regulating redox status of the CXXC motif thereby facilitating copper binding by ATOX1 (99). Copper transporter ATP7A/ATP7B (already described in Fig1) has six conserved CXXC motif located in the N-terminus and one in catalytic site (Transmembrane segment 6). N-terminal CXXC of ATP7B are glutathionylated and that Glutaredoxin1 (GRX1) is responsible for catalyzing deglutathionylation thereby enabling copper binding (96).

Fig 5 – Consequences of the change in the interplay of copper and redox homeostasis in Menkes Disease Fibroblast cells. In wild type cells (Left-Handed Section), copper binds to glutathione after its entry into the cell. Next copper binds to designated chaperones responsible for delivering it to several intracellular locations. Excess copper is exported out of the cells by ATP7A/ATP7B. This export pathway in Menkes Disease Fibroblast is inactivated due to mutation in ATP7A (Right-Handed Section, dotted red arrow). This causes

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copper accumulation in the cytosol and redistribution of excess copper to other intracellular organelles like mitochondria and nucleus (solid red arrow). Excess copper specifically affects glutathione balance in the mitochondria. Redox changes in the mitochondria are signaled to the nucleus (purple dotted arrow) resulting into changes in the gene expression machinery.

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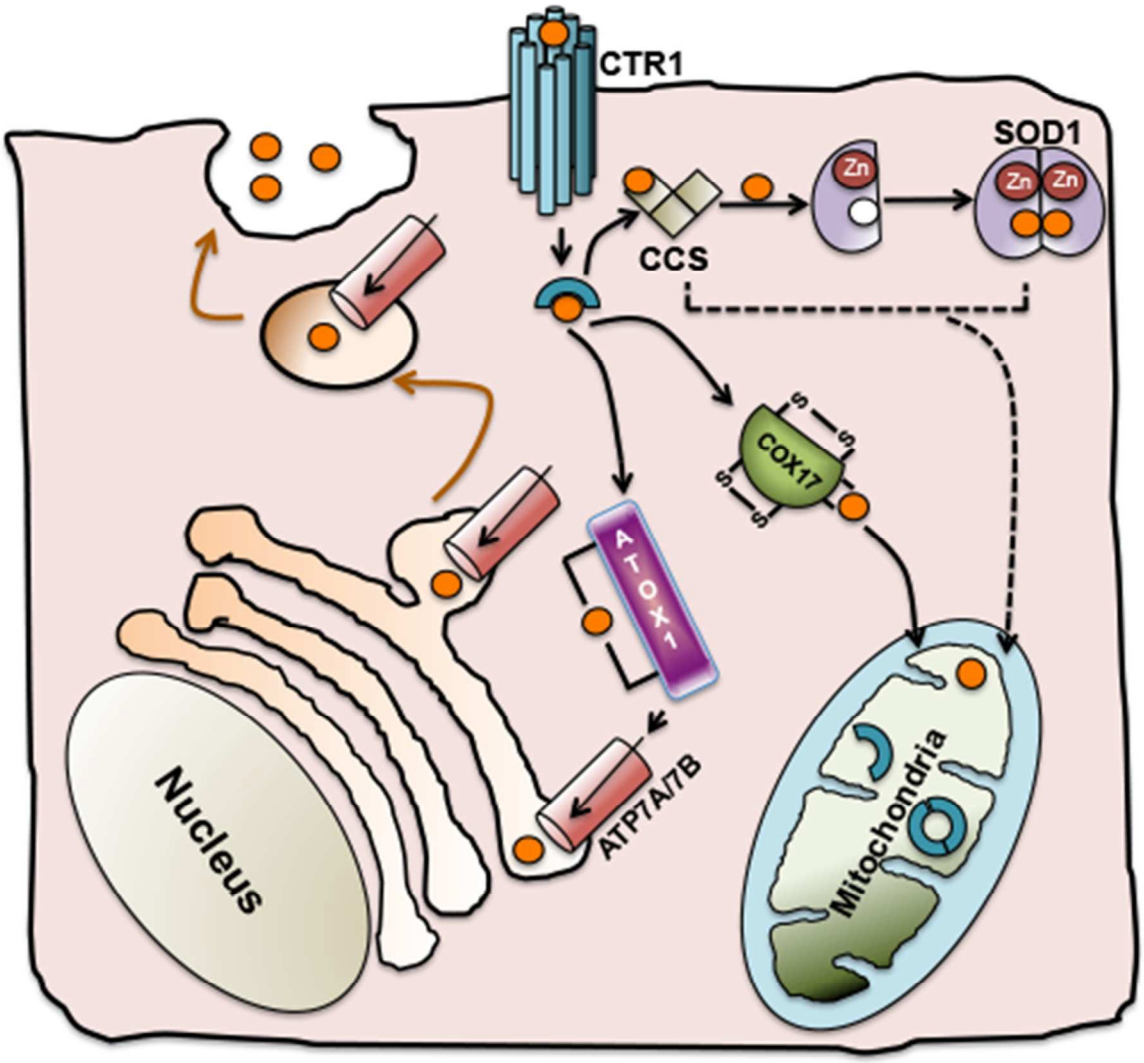


Fig1

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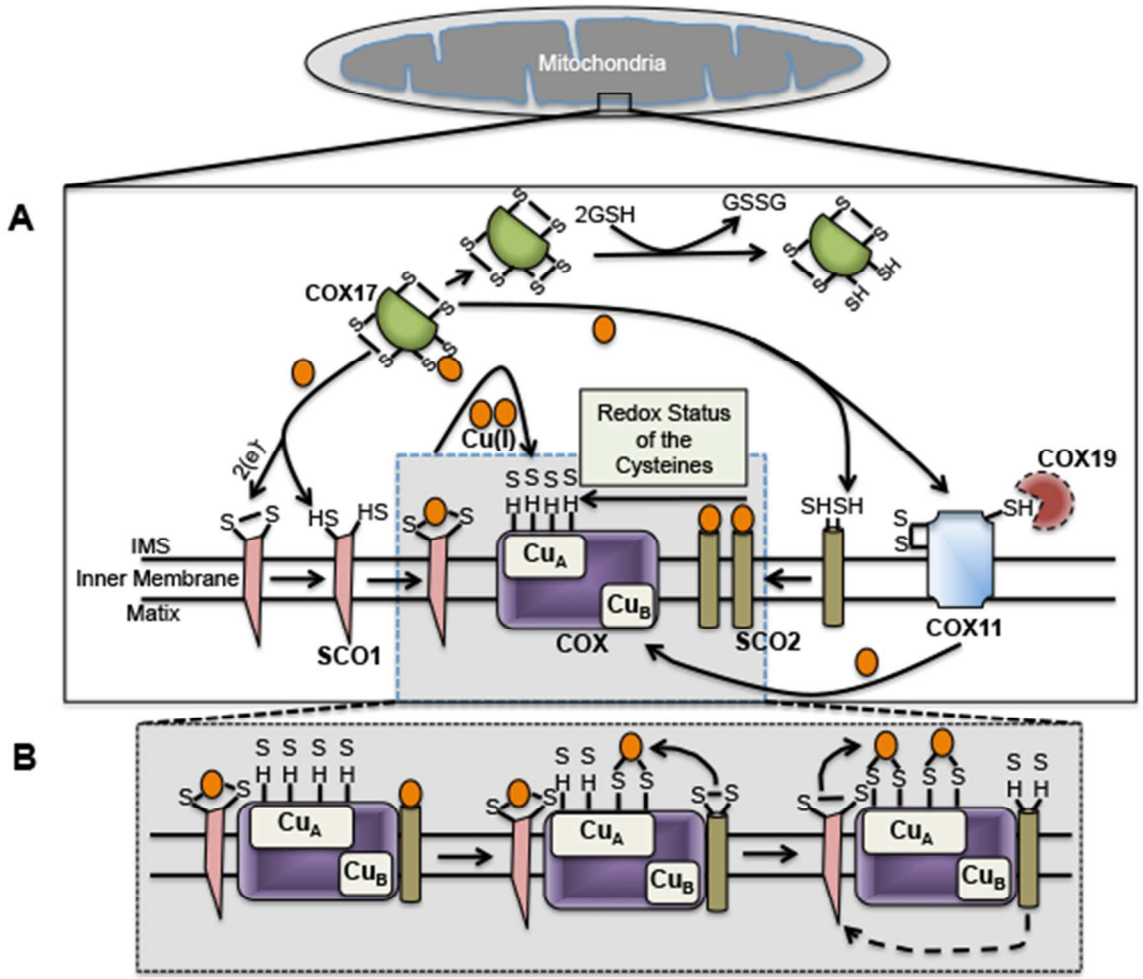


Fig2

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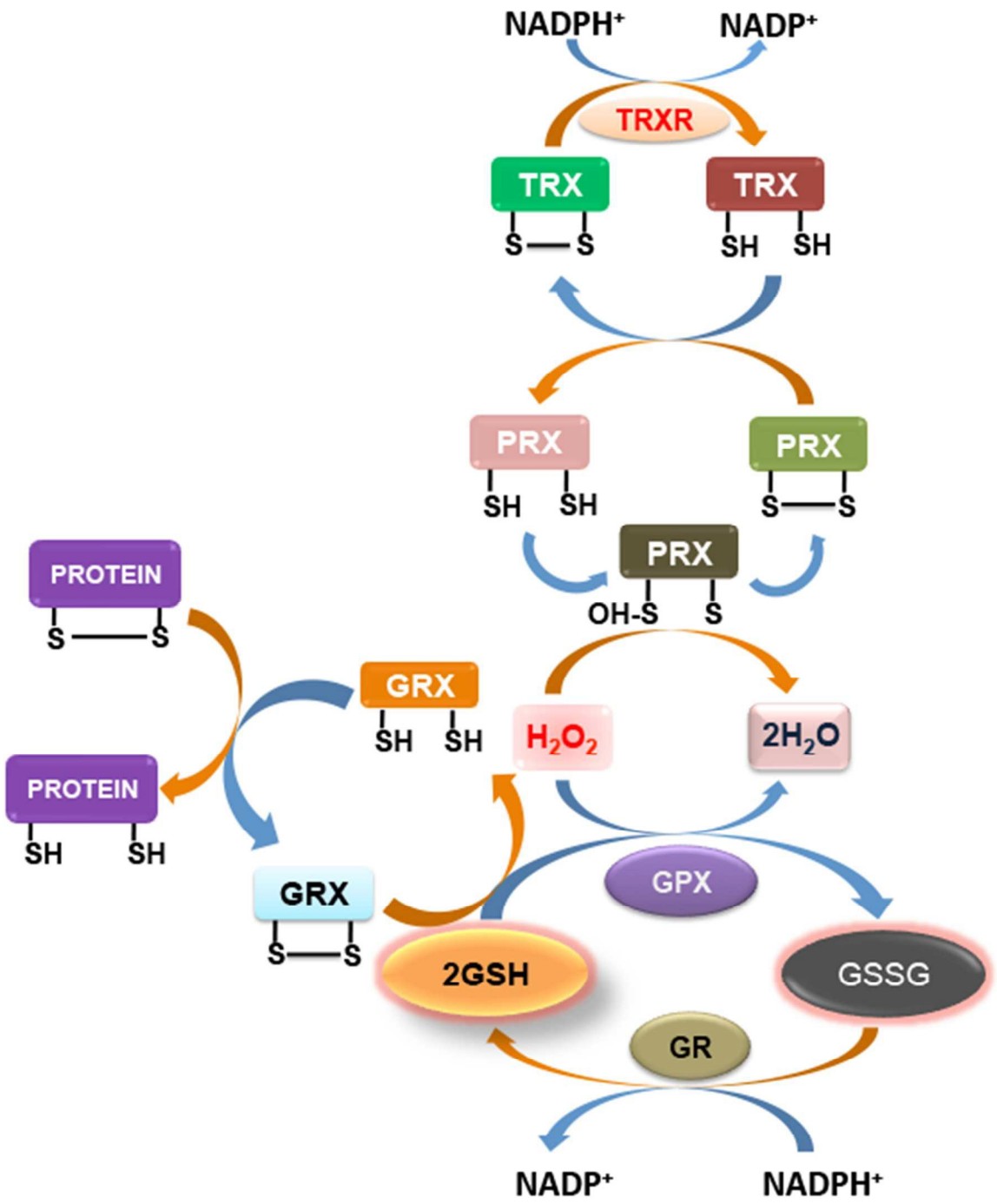


Fig3

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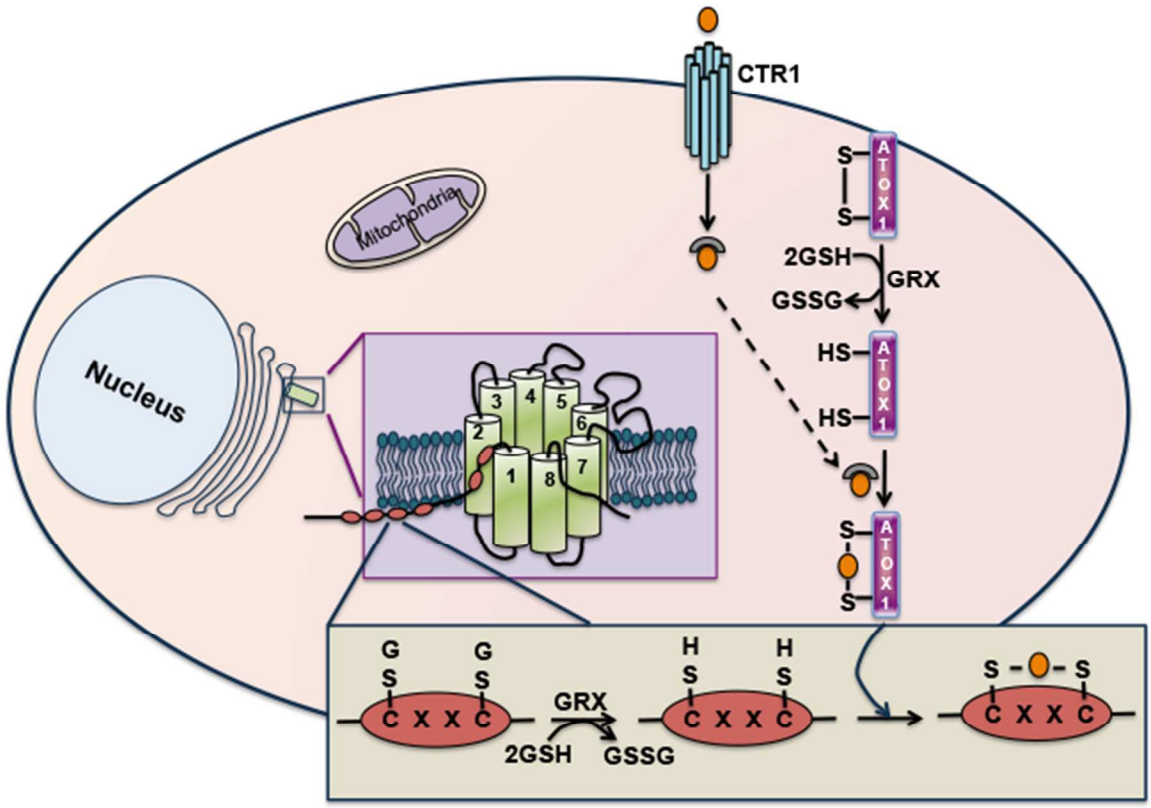


Fig4

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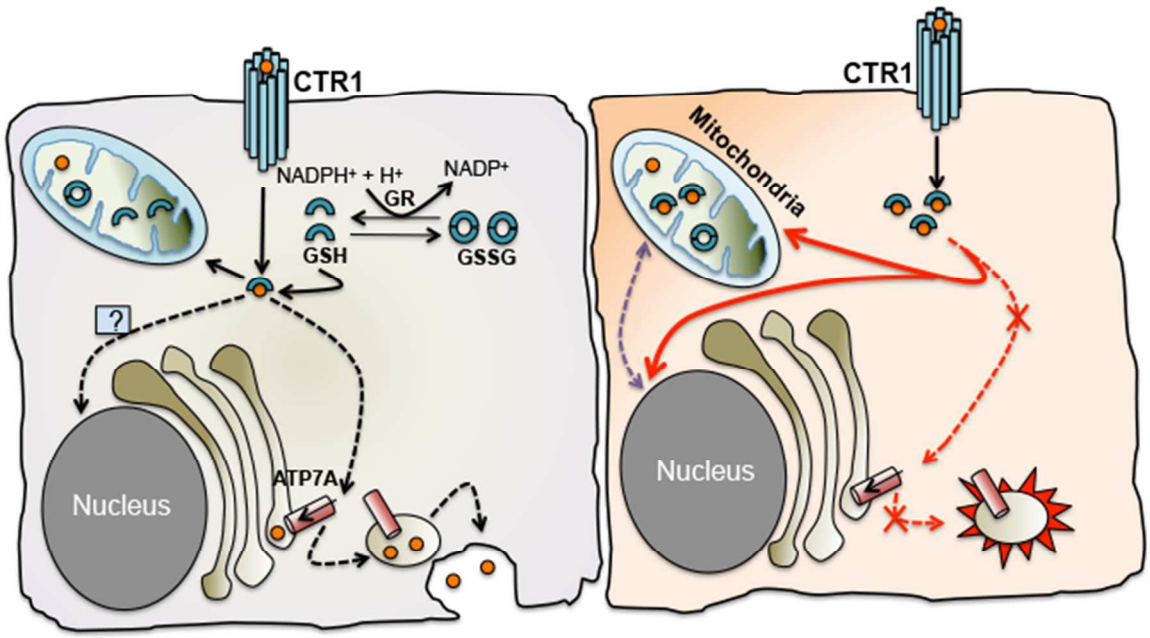


Fig5

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